

BUSINESS ANALYTICS

23CSE452

ANUPAMA NAIR

CB.SC.U4CSE23207

1. Problem Statement and Objectives

1.1 Problem Statement

Recruitment platforms contain a large number of job listings, but these listings do not attract the same level of candidate interest. Candidate engagement can vary based on factors such as job title, company, location, employment type, experience requirements, industry, and the language used in the job description.

This case study focuses on predicting high-performing job listings using Business Analytics and Natural Language Processing (NLP). LinkedIn job-listing data was collected through web scraping using Apify, with candidate application activity used as the performance signal for classifying listings as high-performing or low-performing.

The study investigates whether the textual content of job descriptions can be transformed using TF-IDF (Term Frequency–Inverse Document Frequency) and used with Logistic Regression to predict whether a job listing is classified as high-performing.

Central Business Question:

Which job-listing characteristics are associated with higher candidate interest, and can the text of a job description be used to predict whether a listing will be classified as high-performing?

1.2 Objectives

- To collect and prepare LinkedIn job-listing data through Apify web scraping and perform the required data cleaning and preprocessing.
- To analyze job-listing and job-description characteristics associated with candidate interest using exploratory data analysis.
- To develop and evaluate a TF-IDF and Logistic Regression model for predicting high-performing job listings using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

2. Data Collection and Dataset Description

2.1 Data Collection

The dataset was collected from LinkedIn job postings using Apify, a web-scraping and automation platform. Relevant job roles and search inputs were defined, the LinkedIn job listings were scraped through Apify, and the collected records were exported in CSV format for analysis.

The study covered 18 target roles across technology, analytics, business, engineering, and management, including Data Analyst, Data Scientist, Business Analyst, Data Engineer, Software Engineer, Machine Learning Engineer, AI Engineer, Cloud Engineer, DevOps Engineer, Cybersecurity Analyst, Product Manager, Financial Analyst, Marketing Analyst, HR Analyst, Operations Analyst, Sales Analyst, Project Manager, and UI/UX Designer.

Data Collection Workflow:

LinkedIn Job Search → Define Target Roles → Apify Scraper → Collect Job Listings → Export CSV → Raw Dataset

2.2 Dataset Description

The final raw dataset contains 10,000 job-listing records and 49 attributes. It includes information related to the job, company, location, application activity, salary, and publication details.

2.3 Important Variables

Variable	Description
id	Job-listing identifier
title	Job title
companyName	Hiring company
location	Job location
description	Full job description
applicationsCount	Application-count information
experienceLevel	Required experience level
contractType	Employment/contract type
jobFunction	Functional area of the job
sector	Industry/sector
companySize	Company-size category
salary	Available salary information
publishedAt	Job publication timestamp
high_performing	Binary performance target

The dataset therefore combines structured job and company attributes with unstructured job-description text, supporting both Business Analytics and NLP-based analysis.

3. Data Preparation and Exploratory Analysis

3.1 Data Preparation

The raw LinkedIn job-listing data was cleaned and prepared for exploratory analysis and predictive modelling. Duplicate listings were identified using job title, company name, and location, and duplicate records were removed. Unnecessary scraper-specific metadata, including poster information, scraper indexes, HTML descriptions, company logos, and URLs, was also removed.

The cleaned dataset was then prepared for text-based analysis while taking steps to avoid target leakage.

3.2 Construction of the Performance Target

Candidate application activity was used as the performance signal for each job listing. The application information included both exact numeric counts and descriptions such as “Over 200 applicants” and “Be among the first 25 applicants.”

The application information was categorized into:

- Censored high-count listings
- Censored low-count listings
- Exact numeric application counts
- Unknown values

A binary high_performing target was then created:

Value	Category
0	Low-performing
1	High-performing

Since application activity was used to construct the target, the application-count variables were excluded from the predictive text features. This helped prevent target leakage and ensured that the model did not directly use the information defining the target.

3.3 Text Feature Engineering

Three descriptive features were derived from the job-description text for exploratory analysis:

Feature	Description
Description Length	Number of characters in the job description
Word Count	Number of words in the job description
Sentence Count	Approximate number of sentences in the job description

These features provided basic measures of the amount of textual information contained in each job listing.

3.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted before predictive modelling to understand the structure and characteristics of the dataset.

The analysis included:

- Dataset dimensions and column information
- Data types
- Missing values
- Duplicate records
- Duplicate identifiers

- Distribution of the target variable
- Job-description length
- Job-description word count

The EDA focused particularly on whether basic characteristics of job-description text differed between high-performing and low-performing listings.

3.5 Target Distribution

A bar chart was used to visualize the distribution of the two performance classes: high-performing and low-performing job listings.

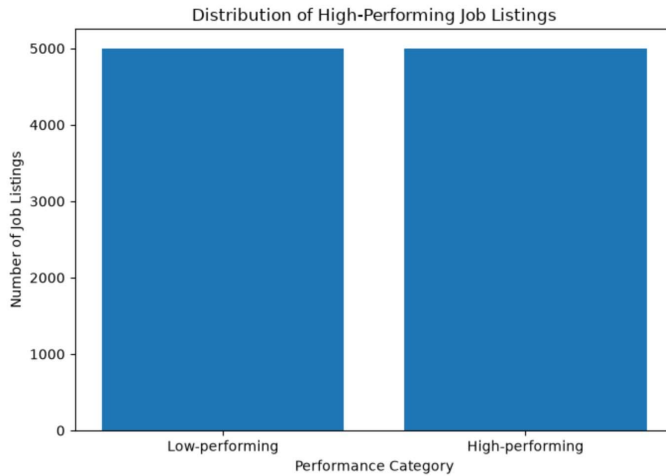


Figure 1. Distribution of job listings according to the high-performing target.

The distribution is approximately balanced between the two performance classes, with similar numbers of low-performing and high-performing listings. This balance reduces the risk of the classifier being dominated by one target class and provides an appropriate basis for evaluating classification performance.

3.6 Job-Description Length Analysis

The distribution of job-description length was examined across the two performance categories using a histogram/KDE visualization.

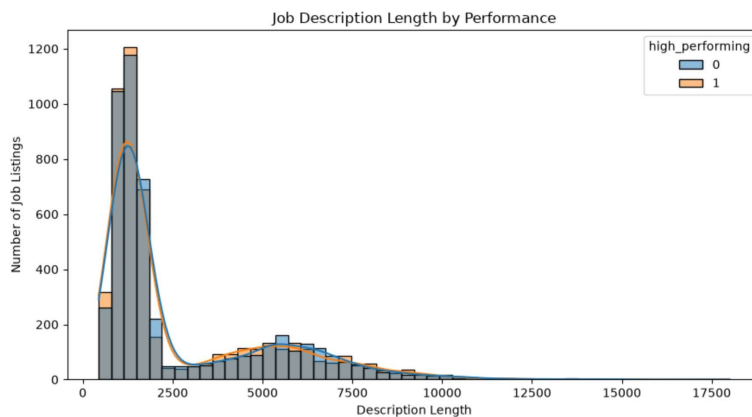


Figure 2. Distribution of job-description length for high performing and low-performing listings.

The distribution shows that most job descriptions are concentrated at shorter lengths, while a smaller number of listings have substantially longer descriptions, producing a long right tail. The two performance categories also show considerable overlap, indicating that description length alone does not clearly distinguish high-performing from low-performing listings.

3.7 Word Count Analysis

A box plot was used to compare the word counts of job descriptions across high-performing and low-performing listings.

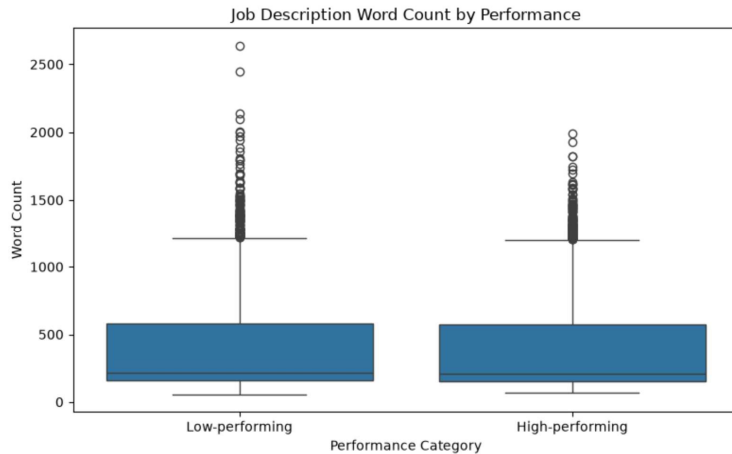


Figure 3. Comparison of jobdescription word counts between high-performing and lowperforming listings.

The word-count distributions of the two categories also show substantial overlap. Both groups contain unusually long descriptions, as indicated by the outliers. This suggests that word count by itself is unlikely to be sufficient for distinguishing listing performance and supports the use of TF-IDF features to capture more specific textual patterns.

3.8 Preparation for NLP Modelling

After preprocessing and exploratory analysis, the dataset used for NLP modelling contained 10,000 records and three variables: id, description, and high_performing. The description field served as the primary textual input, while high_performing served as the classification target.

The prepared job descriptions were subsequently passed to the TF-IDF vectorization stage, where the unstructured text was transformed into numerical features suitable for machine-learning classification.

4. Analytics Method and Implementation

4.1 Analytical Method

The study uses Natural Language Processing (NLP) with TF-IDF and Logistic Regression to classify job listings as high-performing or low-performing. Since the target variable has two classes, the problem is formulated as a binary classification task.

TF-IDF was selected to convert unstructured job-description text into numerical features, while Logistic Regression was used to learn the relationship between these textual features and the performance category. This provides a relatively simple and interpretable approach for analysing job-description language.

4.2 Justification for Method Selection

TF-IDF and Logistic Regression were selected because the primary predictive input is unstructured job-description text and the target variable is binary. TF-IDF provides a computationally efficient representation of important words and phrases, while Logistic Regression is suitable for binary classification and provides interpretable coefficients. The combination therefore provides a balance between predictive capability, simplicity, and interpretability for analysing job-listing performance.

4.3 TF-IDF Feature Representation

Term Frequency–Inverse Document Frequency (TF-IDF) was used to represent the job descriptions numerically. It assigns greater importance to terms that are distinctive within a document while reducing the importance of terms that occur frequently across many documents.

The model considered both unigrams and bigrams, allowing it to capture individual words as well as meaningful two-word phrases.

The main TF-IDF settings were:

Parameter	Configuration
Lowercase conversion	Enabled
Accent normalization	Unicode
N-grams	Unigrams and bigrams
Minimum document frequency	3
Maximum document frequency	0.95
Maximum features	20,000
Term-frequency scaling	Sublinear

These settings helped control the vocabulary size while retaining informative textual patterns.

4.4 Train-Test Split

The dataset was divided into 80% training data and 20% testing data. The training set was used to learn the TF-IDF representation and classification model, while the test set was reserved for evaluating performance on unseen job listings.

A fixed random state of 42 was used for reproducibility, and stratified sampling was applied to maintain the relative distribution of the two target classes.

4.5 Logistic Regression Classification

Logistic Regression was used as the classification algorithm after TF-IDF transformation. The model predicts whether a job listing belongs to:

- Class 0: Low-performing
- Class 1: High-performing

The model was configured with a maximum of 2,000 iterations and a regularization parameter of 1.0. It also generates probabilities for the high-performing class, which are used in ROC-AUC evaluation.

The TF-IDF transformation and Logistic Regression classifier were implemented as a single pipeline to ensure that the text vocabulary was learned from the training data as part of the modelling process.

4.6 Model Evaluation

The model was evaluated using multiple performance measures:

Metric	Purpose
Accuracy	Measures the overall proportion of correctly classified listings.
Precision	Measures how many listings predicted as high-performing were actually high-performing.
Recall	Measures how many actual high-performing listings were correctly identified.
F1-Score	Provides a combined measure of precision and recall.
ROC-AUC	Measures the model's ability to distinguish between the two performance classes across different thresholds.

A classification report, confusion matrix, and ROC curve were also used to provide a more detailed evaluation of the model.

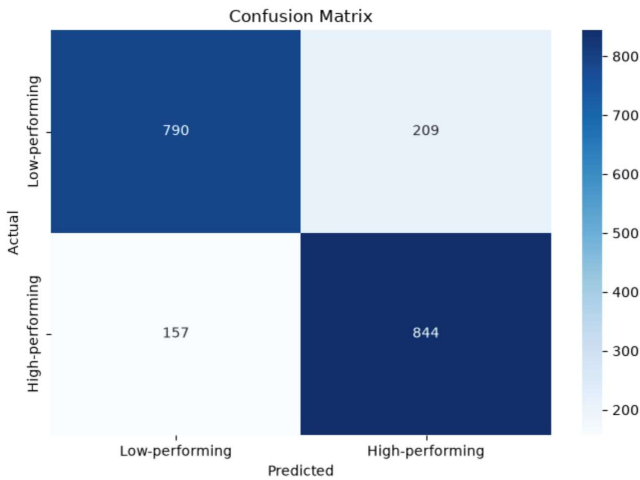


Figure 4. Confusion matrix of the classification model.

The confusion matrix shows the numbers of correctly and incorrectly classified highperforming and low-performing job listings, providing a clear view of the model's classification errors.

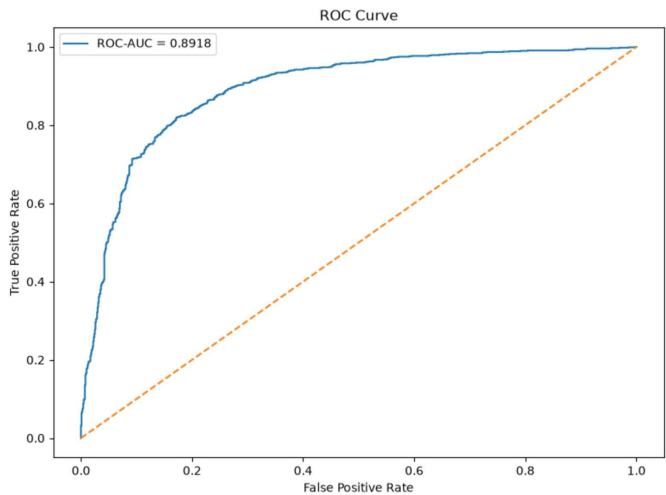


Figure 5. ROC curve of the classification model.

The ROC curve illustrates the model's ability to distinguish between high-performing and low-performing listings across different classification thresholds. A larger area under the curve indicates stronger class-separation ability.

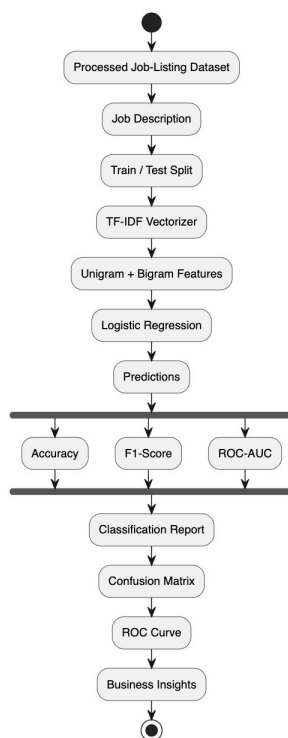
4.7 Interpretation of Textual Features

The Logistic Regression model also provides an interpretable view of the textual features through its TF-IDF coefficients. Terms with larger positive coefficients are more strongly associated with the high-performing class, while terms with negative coefficients are associated with the low-performing class.

These relationships represent statistical associations rather than causal effects. Therefore, the presence of a particular word or phrase should not be interpreted as directly causing higher candidate interest.

4.8 Overall Implementation

The overall analytical process can be summarized as:



The methodology therefore provides an end-to-end analytical process for transforming unstructured job-description text into measurable features and using those features to classify job listings according to the defined performance target.

5. Comparison with State-of-the-Art Methods

Recent research has used machine learning and text analysis to study job advertisements, vacancy attractiveness, and applicant behaviour. The following studies provide relevant comparisons with the proposed TF-IDF and Logistic Regression approach.

(Refer to the table below)

Published Study / Year	Dataset	Method Used	Evaluation Metric / Approach	Key Result	Comparison with Your Work
Košťálová, Lyócsa & Štefánik (2022)	32,482 online job vacancies	Linear models and Random Forest with job-ad and text- related features	MSE and MAE	Examined factors associated with online vacancy attractiveness, including views, reactions and applications.	The study examines job-advertisement attractiveness using a different dataset size, prediction target, and modelling approach.
Frissen, Adebayo & Nanda (2023)	Employment Scam Aegean Dataset containing real-life job advertisements	Word embeddings, BERT and several ML classifiers including Logistic Regression and Random Forest	Accuracy, Precision, Recall and F1-score	Demonstrated the use of NLP and ML for identifying bias and discrimination in job advertisements.	Both studies use NLP and classification of job-ad text. However, the present study predicts job-listing performance rather than bias or discrimination.
Chaturvedi, Mahajan & Siddique (2025)	157,888 Indian job advertisements with 6.45 million applications	Text analysis and regression based analysis	Regression and Applicant behaviour analysis	Found that language in job advertisements was associated with applicant behaviour and gender-related application patterns. (IDEAS/RePEc)	Both studies examine the relationship between job-ad text and applicant behaviour. The present study uses 10,000 LinkedIn listings and performs direct binary classification of listing performance.
This Study	10,000 LinkedIn job listings	TF-IDF + Logistic Regression	Accuracy, Precision, Recall, F1score and ROC-AUC	Predicts whether a job listing is classified as high performing using its job-description text.	Focuses specifically on predicting high-performing listings using an interpretable text-classification approach.

5.1 Comparison

The reviewed studies show that job-advertisement text can provide useful information for analysing vacancy attractiveness, applicant behaviour, and other recruitment outcomes.

The present study differs mainly in its target variable and modelling objective. It uses TF-IDF and Logistic Regression to directly classify job listings as high-performing or low-performing based on their descriptions. The approach is relatively simple and interpretable, while the compared studies use different datasets, objectives, and analytical methods. Therefore, their reported results are not directly comparable as performance scores.

5.2 Strengths and Limitations of the Compared Approaches

The reviewed studies use different datasets and analytical objectives, ranging from vacancy attractiveness to bias detection and applicant behaviour. More complex NLP approaches can capture richer linguistic patterns, while simpler approaches such as TF-IDF provide greater interpretability and are easier to implement. A limitation of the present study is that the prediction is based primarily on job-description text, whereas job performance may also depend on factors such as company characteristics, location, salary, role type, and market conditions.

6. Results, Business Insights and Recommendations

6.1 Results

The TF-IDF and Logistic Regression model was evaluated on the test dataset using multiple classification metrics.

Metric	Score
Accuracy	81.70%
Precision	80.15%
Recall	84.32%
F1-Score	82.18%
ROC-AUC	0.8918

The model achieved an accuracy of 81.70%, while the recall of 84.32% indicates that a substantial proportion of highperforming job listings were correctly identified. The ROC-AUC of 0.8918 indicates good separation between the two performance classes.

6.2 Interpretation of Model Performance

The model achieved an accuracy of 81.70%, indicating that a substantial proportion of job listings were correctly classified. The precision of 80.15% shows that most listings predicted as high-performing were correctly identified, while the recall of 84.32% indicates that the model successfully identified a large proportion of the actual highperforming listings. The F1-score of 82.18% demonstrates a balanced performance between precision and recall. The ROC-AUC of 0.8918 further indicates that the model has good ability to distinguish between high-performing and lowperforming job listings.

Confusion Matrix Error Analysis: The confusion matrix for the 2,000-record test set shows that 790 low-performing listings and 844 high-performing listings were correctly classified. The model incorrectly classified 209 low-performing listings as high-performing and 157 high-performing listings as low-performing. The relatively higher number of correctly identified high-performing listings is consistent with the recall of 84.32%. These errors indicate that jobdescription text alone cannot perfectly capture all factors influencing listing performance.

6.3 Business Insights

- Job-description text contains useful predictive information for identifying high-performing job listings.
- The model can help recruiters identify listings that are more likely to attract higher candidate interest.
- The TF-IDF features provide interpretable textual patterns associated with the performance categories.
- The identified relationships are associations rather than causal effects; a particular word or phrase does not necessarily cause higher candidate interest.

6.4 Recommendations

1. Use NLP as a decision-support tool: Recruiters can use the model to screen job descriptions and identify potentially high-performing listings before publication.
2. Improve job-description quality: Job descriptions should be clear, relevant, and informative to improve the overall quality of job postings.
3. Use textual insights for refinement: Patterns identified through the NLP analysis can support the review and refinement of job descriptions.
4. Retrain periodically: The model should be evaluated and updated using newer job-listing data to account for changes in recruitment trends and language.
5. Retain human oversight: Model predictions should support recruiter decisions rather than replace human judgement.

6.5 Study Limitations

The study has several limitations. First, the performance target is derived from application activity, which may not capture all aspects of candidate interest or job quality. Second, the predictive model primarily uses job-description text and therefore does not directly incorporate potentially relevant factors such as salary, company characteristics, location, and role-specific conditions. Third, the dataset represents the job listings collected during the study period and may therefore reflect recruitment patterns specific to that period. These limitations should be considered when applying the model to other job markets or future datasets.

6.6 Future Scope

Future work can extend the model by incorporating structured job attributes such as salary, experience level, employment type, location, company size, and industry alongside job-description text. More advanced NLP approaches such as transformer-based language models could also be evaluated. Additional data collected across different time periods and job markets could help assess how well the model generalizes to changing recruitment trends.

7. Conclusion and References

7.1 Conclusion

This case study analysed 10,000 LinkedIn job listings to investigate the characteristics associated with higher candidate interest and to determine whether job-description text can be used to predict high-performing listings.

The study combined Business Analytics and Natural Language Processing by using TF-IDF to transform jobdescription text into numerical features and Logistic Regression for binary classification. The model achieved 81.70% accuracy, 82.18% F1-score, and 0.8918 ROC-AUC, demonstrating that job-description text contains useful information for distinguishing between high-performing and low-performing listings.

The findings indicate that NLP-based analysis can provide useful decision-support for recruiters when reviewing job descriptions. However, the identified textual patterns represent associations rather than causal relationships, and model predictions should be used alongside human judgement. Future work could incorporate additional job and company attributes and evaluate the model on newer job-listing data.

8. References

1. Košťálová, Z., Lyócsa, Š., & Štefánik, M. (2022). Online job vacancy attractiveness: Increasing views, reactions and conversions. *Electronic Commerce Research and Applications*, 55, 101192. <https://doi.org/10.1016/j.elerap.2022.101192>
2. Frissen, R., Adebayo, K. J., & Nanda, R. (2023). A machine learning approach to recognize bias and discrimination in job advertisements. *AI & Society*, 38, 1025–1038. <https://doi.org/10.1007/s00146-022-01574-0>
3. Chaturvedi, S., Mahajan, K., & Siddique, Z. (2025). Gendered language in job ads and applicant behavior: Evidence from India. *Labour Economics*, 96, 102726. <https://doi.org/10.1016/j.labeco.2025.102726>
4. LinkedIn. (n.d.). *LinkedIn Jobs*. <https://www.linkedin.com/jobs/>
5. Apify. (n.d.). *Apify: Web scraping and automation platform*. <https://apify.com/>
6. Scikit-learn developers. (n.d.). *TfidfVectorizer documentation*. https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html